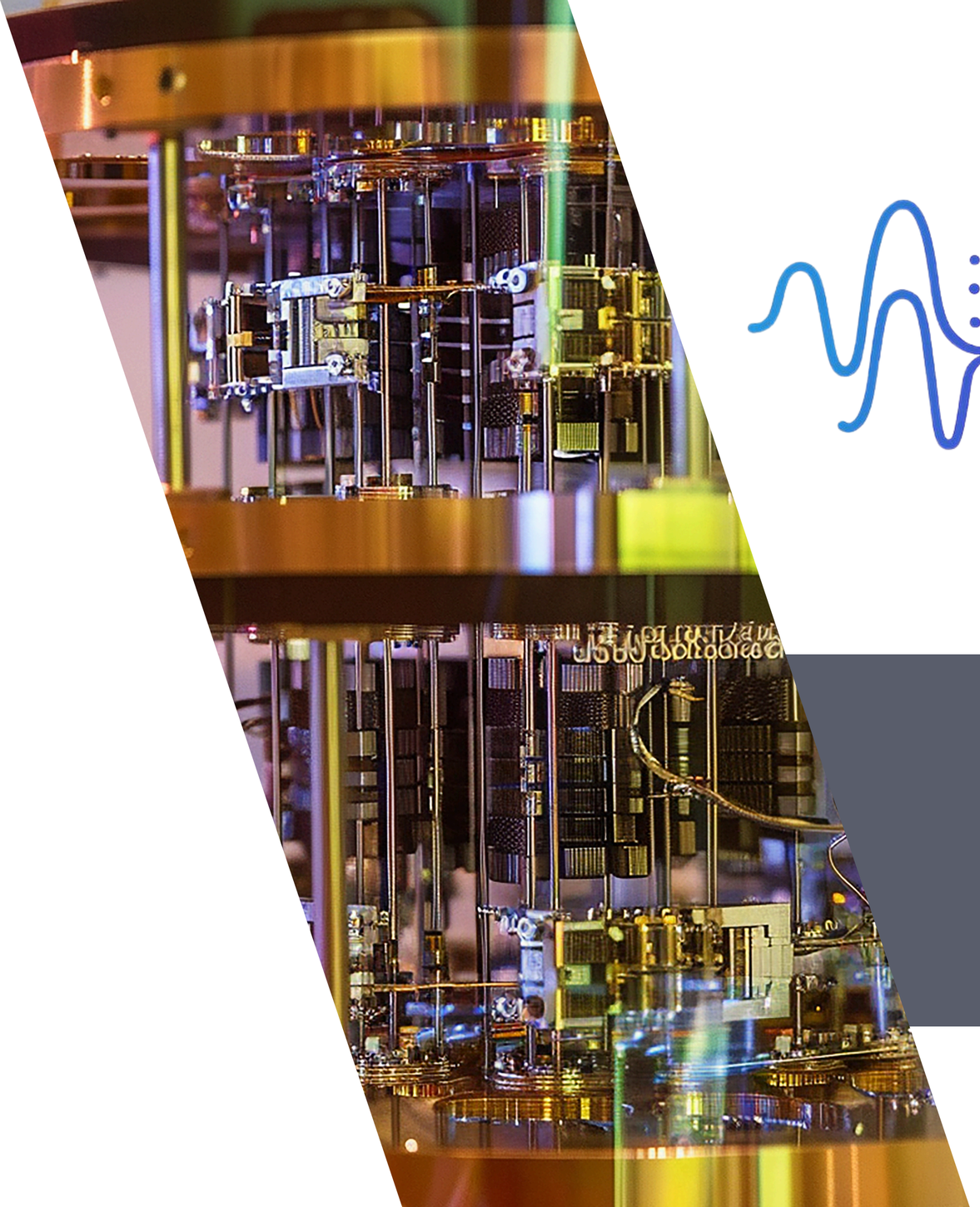
Natural Language → Quantum Code for Drug Discovery

Transforming Quantum Computing



SPECTRIX

BioQL

Turning Words Into Quantum Drug Discovery

The problem:

Slow / Expensive

Drug discovery is slow (10–15 years) and expensive ($2B+ per drug).

Too Technical

Quantum tools are too technical (Qiskit, Cirq) for most teams.

Lack Expertise

Pharma and biotech lack in-house quantum expertise

Challenges & Opportunities in Data Science

THE solution:

01

LLM

LLM-powered NL→QC compiler specialized for biotech.

02

Natural Lenguage

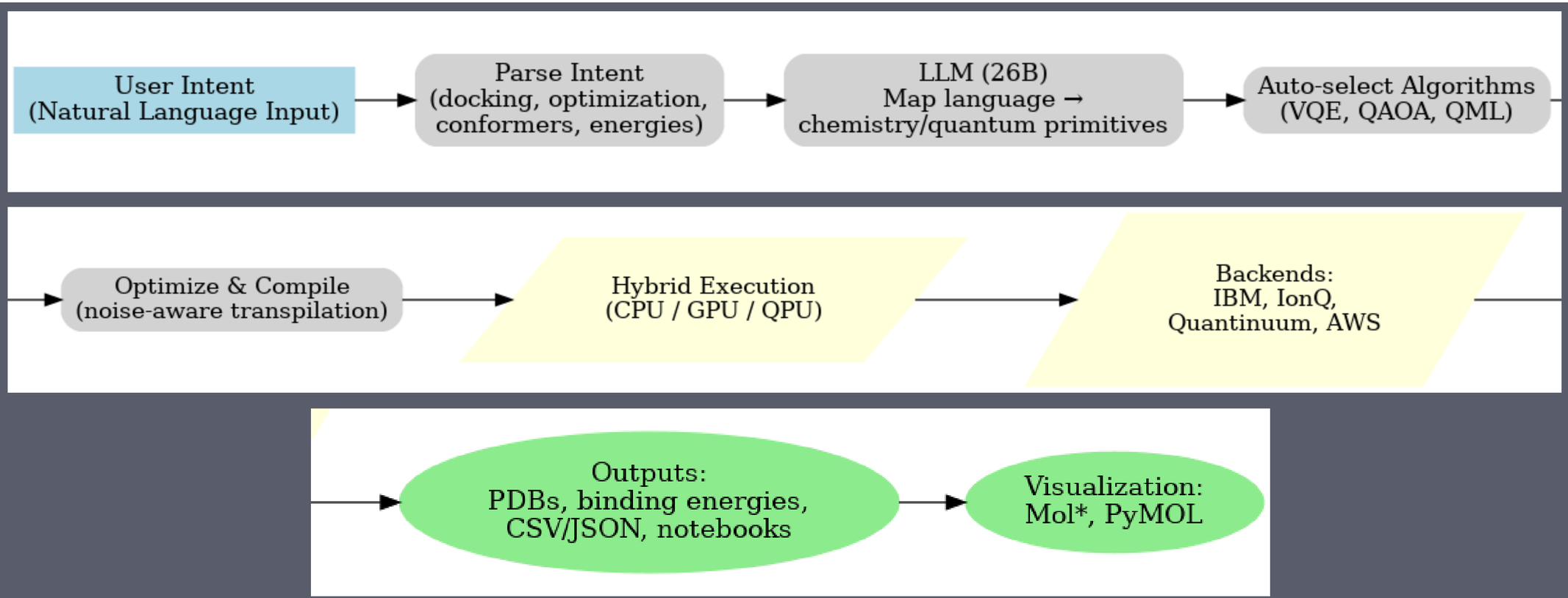
Describe tasks in plain English; get runnable quantum workflows.

03

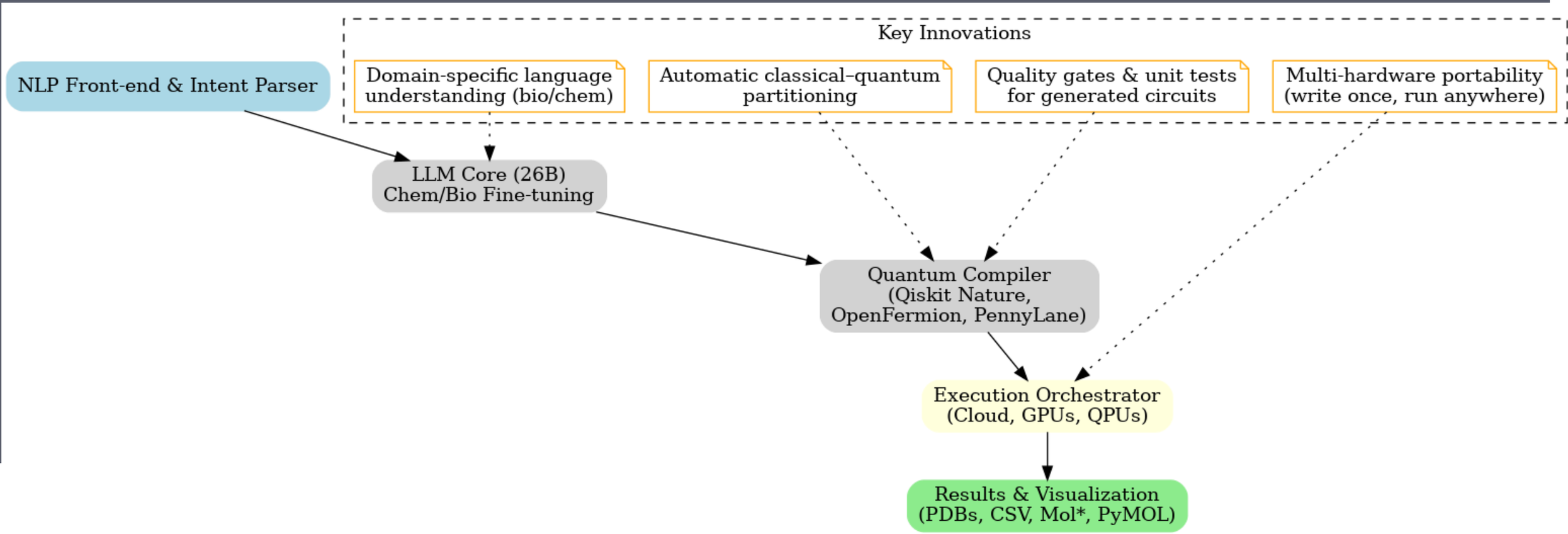
Biotech power

Docking, molecular simulation, PDB generation across IBM/Azure/AWS

How BioQL Works



Architecture (High-Level)



Code Example

```
# BioQL example: docking with hybrid quantum VQE

from bioql import Docking

# Define a docking task: ligand (SMILES) with target protein (PDB)
docking = Docking(
    ligand="CC1=... (SMILES: berberine derivative)",
    protein="GLP1R.pdb",
    method="hybrid_quantum_vqe",              # algorithm: hybrid quantum VQE
    options={
        "basis": "sto-3g",                    # quantum chemistry basis set
        "optimizer": "spsa"                   # optimizer for variational parameters
    }
)

# Run the docking on IBM Quantum backend (simulator in this case)
results = docking.run(backend="ibm_qasm_simulator")

# Output results
print("Binding energy:", results.binding_energy, "kcal/mol")
print("Conformation file:", results.best_pdb)
```

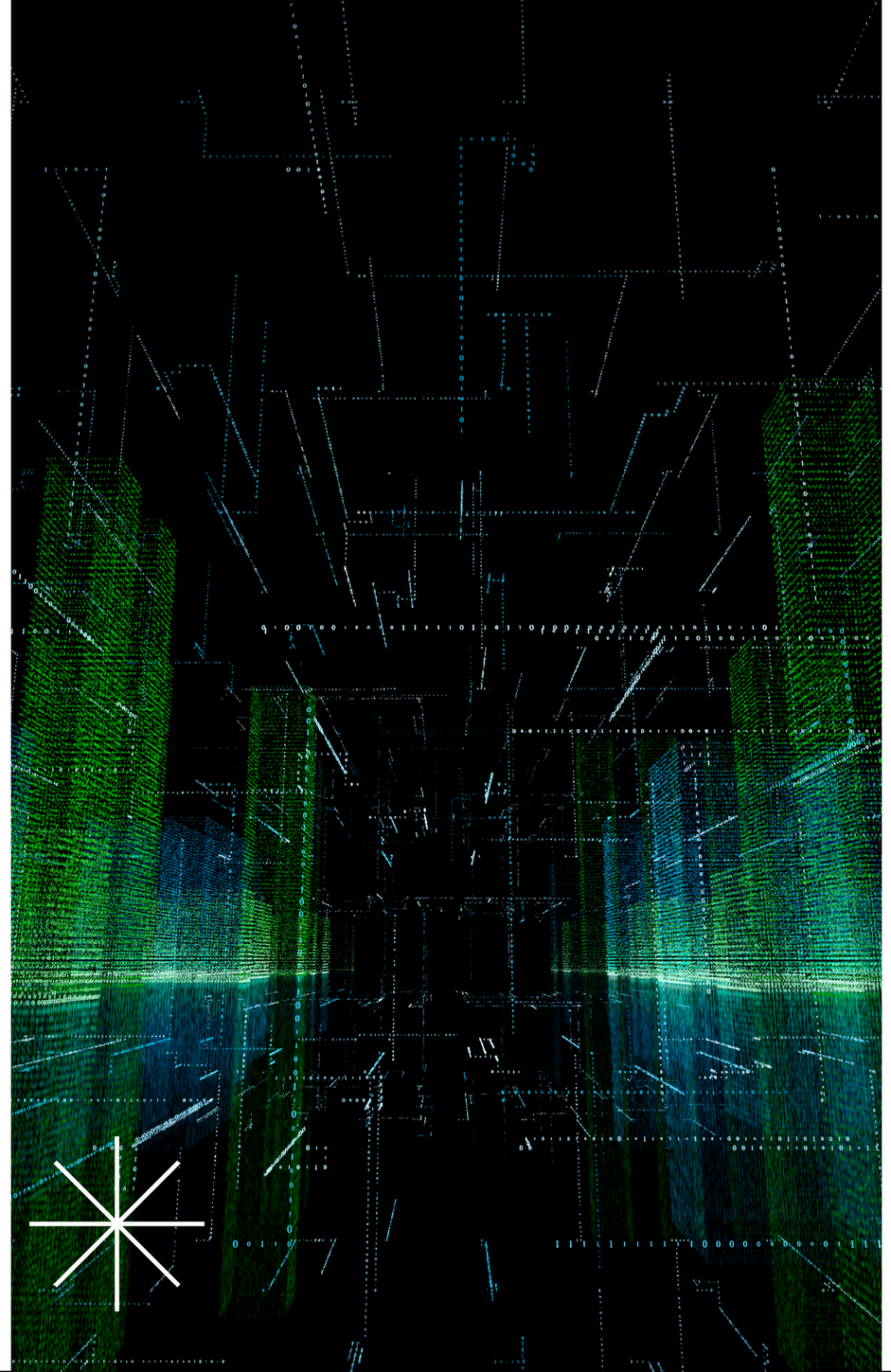
Market & Timing

Why Now

- Quantum hardware accessible via cloud.
- Biotech hungry for acceleration & differentiation.
- LLMs make expert workflows accessible to non-experts

TAM/SAM/SOM (Directional)

- QC software market → multi-$B by 2030.
- Drug discovery software niche → $3B+.
- Early customers: academia & agile biotech teams.



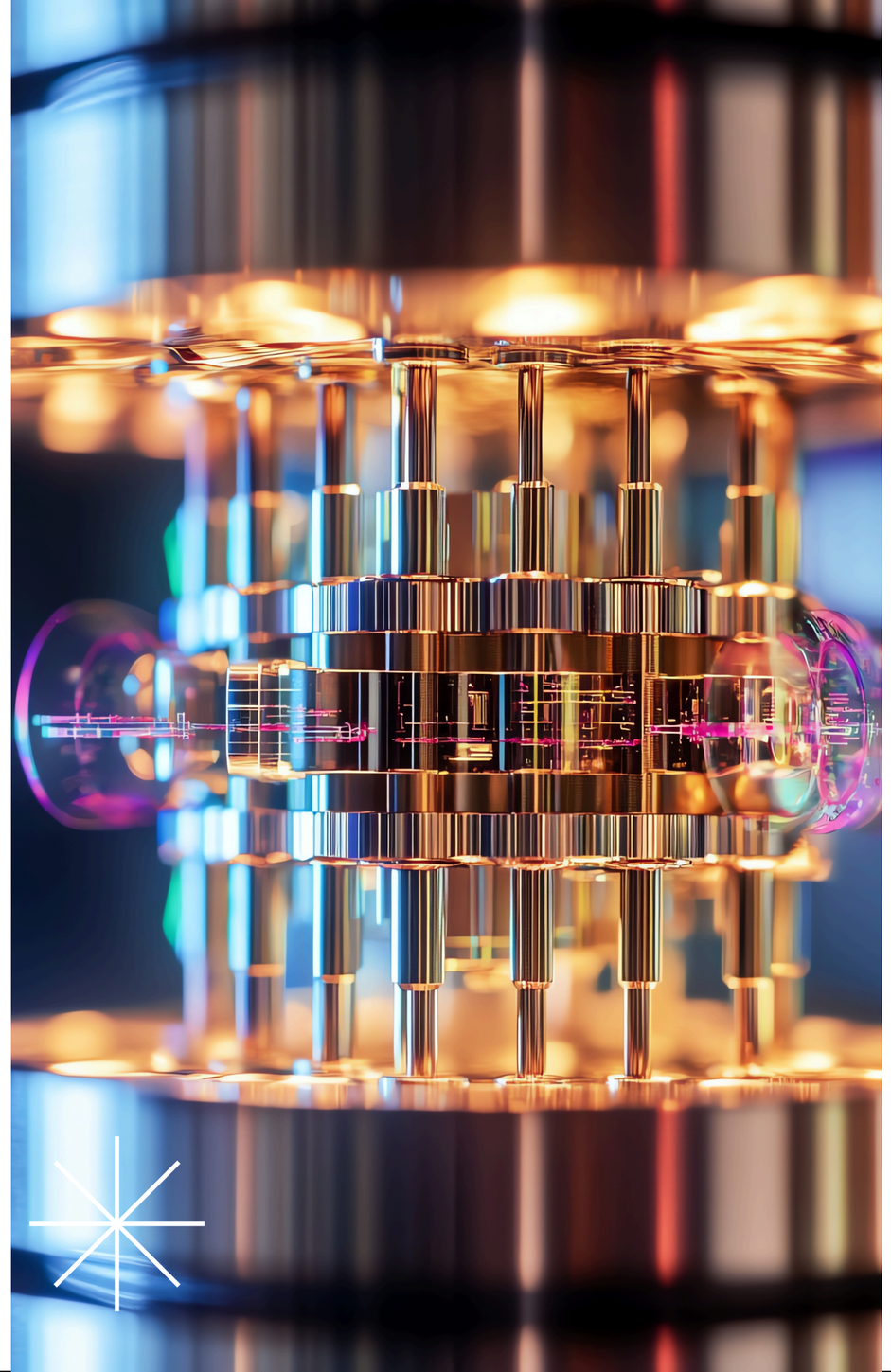
Competition & Positioning

BioQL Edge

- Biotech- first NL→QC compiler (domain- specific).
- End- to- end pipelines (not just circuits).
- Multi-hardware portability.
- Usable by non- quantum experts from day one.

Players

- Zapata (Orquestra), QC Ware (Promethium),
- Qubit Pharma (Atlas), PolarisQB, ProteinQure,
- Quantinuum λambeq (language→circuits, general NLP).



Business Model

Revenue Streams

- SaaS tiers: Academic $149, Biotech $2099, Pharma $4,999+.
- Enterprise licensing & integrations (LIMS/ELN/HPC).
- Cloud marketplace partnerships (IBM/Azure/AWS).

Go- to- Market

- Academic freemium → publications & community.
- Biotech pilots (paid PoCs) → case studies.
- Enterprise pharma → security, SLAs, SOC2.

Funding Plan & ROI

Seed Round (Recommended)

- Raise: $800K for 20% (post $4M).
- Runway: 18 months to break- even.
- Use: hires, pilots, compliance, cloud

Investor Return (3-year)

- Base: 12.5x (to ~$10M equity value).
- Conservative: 7.5x (~$6M).
- IRR range: ~95%–174%.

Risks & Mitigations

Funding risk →	Mitigation
•Bootstrap Plan B (lean). •Hardware pricing → Multi-cloud backends. •Competition → UX moat + domain specialization.	•Market timing → Reduce burn / extend runway. •Churn → Early customer success & NPS loops.

Roadmap & Milestones

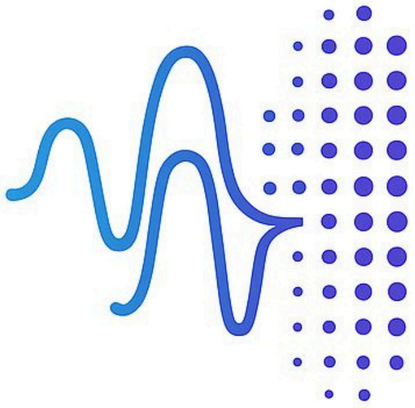
- M6: $5K MRR; 50 paying; 1 biotech.
- M12: $15K MRR; 150 paying; 10 biotech; 1 pharma; SOC2 Type I.
- M18: $80K MRR (break-even).
- M24: $160K MRR; 1,000 paying; 15 pharma; prep Series A.

Founder

Heinz Jungbluth

- Peruvian
- Bsc Biochemistry
- Biotech Founder
- AI-ML expertise

Vision (5–10 Years)



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BioQL

- BioQL becomes the global standard interface between biotech and quantum computing.
- Quantum becomes invisible yet indispensable in pharma pipelines.
- Powering breakthroughs impossible with classical compute.

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